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Text to 3D Urban Scene Generation

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Problem Statement

To develop a text-driven 3D scene generation framework capable of creating large-scale urban environments from natural language descriptions. The system should support realistic modeling of buildings, roads, and public infrastructure while maintaining geometric and view consistency. The goal is to assist smart city planning and visualization by enabling rapid prototyping of urban layouts without manual 3D modeling.

Phase-1 Recap

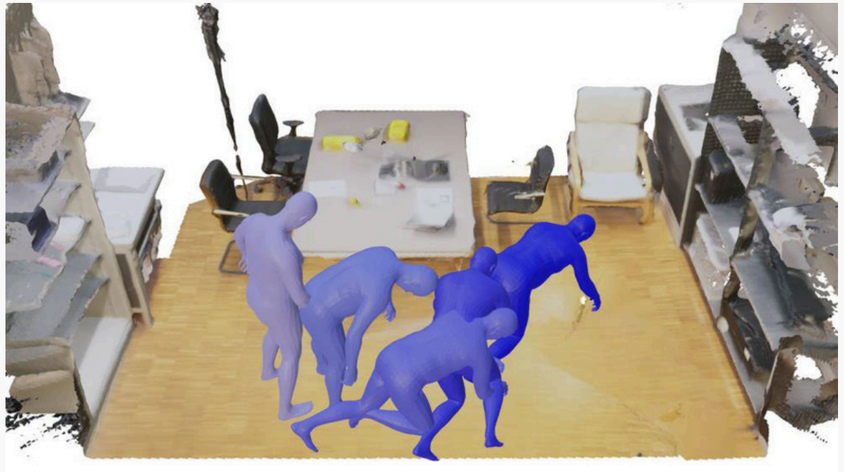
Text-to-NeRF combines text-guided diffusion models with Neural Radiance Fields to generate realistic, view-consistent 3D scenes directly from natural language descriptions, reducing the need for manual 3D modeling.

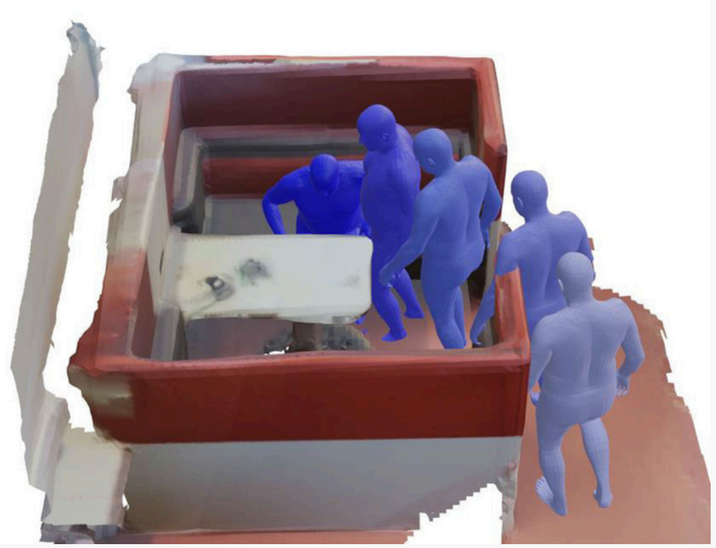
A text-guided diffusion model is a generative AI model that creates images (or other data) by progressively removing noise, where the direction of generation is controlled by a text prompt.

A Neural Radiance Field (NeRF) is a neural network that:

- Takes a 3D point in space and a viewing direction as input
- Outputs:
  - Color (RGB)
  - Volume density (how opaque the point is)

Using these outputs, NeRF can render 2D images from arbitrary camera angles.





Methodology:

- Neural Radiance Fields (NeRF) as 3D Representation
- Text-to-Image Diffusion Model as Content Prior
- Monocular Depth Estimation as Geometric Prior
- PIU(Progressive Inpainting and Upadating)
- Support Set
- Depth-Aware NeRF Optimization

Why Phase-2 was needed?

- Limited Structural and Physical Correctness.
- No Explicit Scene Graph or Object-Level Control.
- Evaluation Limited to Image-Based Metrics.

Literature Survey: Urban Architect(10 April 2024)

Prelimnaries:

- SDS(Score Distillation Sampling)-Improves 3D model by comparing its rendering image with what a diffusion model thinks.
- VSD(Variational Score Distillation)-Generating multiple possible 3D scenes instead of just one.

LGVSD(Layout guided Variational Score Distillation):

- Geometric and Semantic Constraints.

Layout Aware Refinement :

- This model refines details like depth,edges and textures.

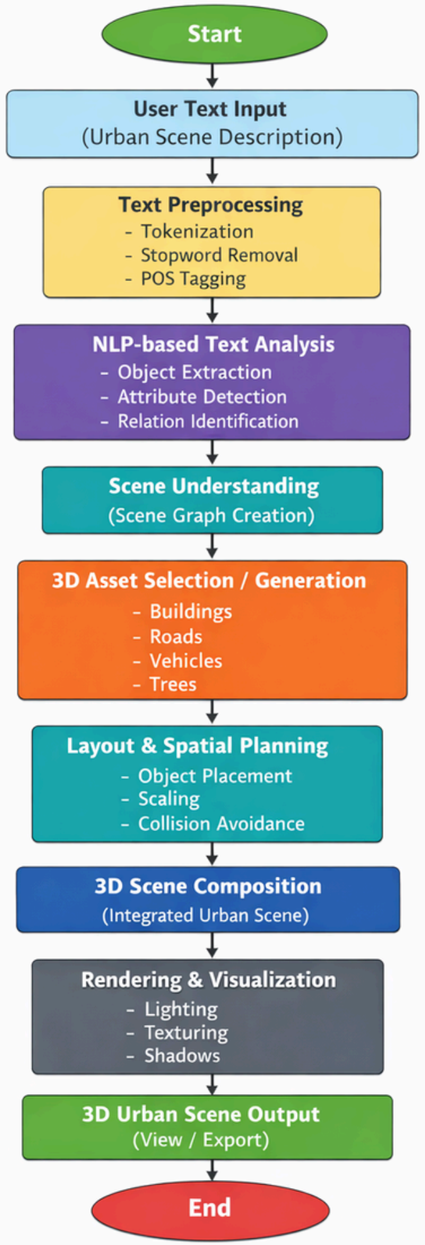
SHG(Scalable Hash Grid) :

- Very large and Unbounded Model

Scene Editing:

- It supports editing like changing styles(night,rainy,cartoon)

Workflow of Text to Urban 3D Scene Generation System



System Architecture:

Text Processing Module:

- Using NLP extracts objects(building,roads,cars,trees),Attributes,Height,Color,Size.
- Techniques Used: Tokenization,NER,Dependency Parsing,Transformer Based Models like BERT.

Scene Understanding Module :

- Converts text understanding into a scene graph.

3D Asset generation Module:

- Generate 3D models for each object.
- Techniques Used: Text-3D Diffusion Models,NeRF,Mesh generation networks.

Layout and Spatial Planning Module:

- It helps in collision avoidance,road alignment,building spacing.
- Algorithms Used: Graph Based Layout Optimization,Grid based Road generation.

System Architecture:

Rendering and Visualisation Module:

- It adds lightning,shadows,textures,whether effects.
- Tools : Blender,OpenGL.

Machine Learning Models :

- NLP Models like BERT,T5,GPT based Text Encoders.
- 3D Generative models like : DreamFusion,Point-E,3D Scene Dreamer,Latent NeRF Model.

Layout Optimization Models:

- GNS(Graph Neural Networks) - Designed to work with graphs.
- Reinforcement Learning for Scene Placement - Automatically deciding where and how to place objects in a scene to get a final realistic look.

Applications:

- Smart City Planning & Visualization
- Urban Infrastructure Design
- Traffic & Transportation Simulation 🚦
- Disaster Management & Risk Assessment 🚨
- Environmental Impact & Sustainability Analysis 🌱
- AR/VR for Urban Exploration 🥽
- Education & Research 🎓

Conclusion

The Text to Urban 3D Scene Generation project demonstrates an intelligent approach to creating urban environments directly from natural language descriptions.It effectively combines NLP, machine learning, and 3D graphics to automate city scene creation.The system interprets textual input, identifies urban elements, and generates structured 3D layouts.This reduces manual modeling effort and speeds up urban visualization.The project is useful for smart city planning, simulations, and virtual environments.With further enhancements, it can achieve higher realism and real-time scene generation.

Thank you